SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Total Marks: 25

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Code of Conduct

I G. Venkata Goutham Reddy (192324265) (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Goutham

Assessment Tasks
Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Total Marks:

ASUS Vivobook

Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

* Cloud computing is the delivery of computing resources such as servers, storage, databases, networking and software over the internet on demand. Users can access these resources without purchasing and maintaining their own physical infrastructure.

Four characteristics:-

1. On-demand self-service:-

* User can create and use resources such as virtual machines and storage whenever required without directly contacting the provider.

2. Board network access:-

* Cloud services can be accessed through the internet using computers, laptops, tablets and smartphones.

3. Resource pooling:-

* Cloud provides share a large pool of computing resources among multiple users. virtualization helps allocate resources effiently.

4. Rapid elasticity:-

* Resources can be quickly increased or decreased according to demand.

Example:- An online shopping website can add more servers during a festival sale

Difference from on-prmise computing:-

* In traditional on-premise computing, an organization purchases and maintains its own hardware and software. cloud computing provides resources on demand, offers easy

scalability, and generally follows a pay-as-you-use model.

Example:- AWS provides computing and storage resources without requiring users to purchase physical servers.

2) Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modren cloud platforms.

cloud computing evolved through several stages of hardware and software development.

- Mainframe computing:- many users shared a large centralized computer through terminals.
- Personal computers:- Computing became available to individual users, but each organization had to manage its own hardware
- computer networks:- computers cloud communicate and share resources.
- client-server computing:- clients accessed applications and data from centralized servers.
- virtualization:- one physical server could run multiple virtual machines, improving resource utilization and flexibility.
- Internet & web:- Applications and services could be accessed from anywhere through the internet.
- modren cloud computing:- virtualization, networking, automation, distributed storage, and data centers were combined to provide computing resources as services.

the evolution can be represented as:-

Main frames → PCs → Networks → client-server → virtualization → Internet/web → cloud computing.

These developments enabled modren cloud platforms such as AWS, Microsoft Azure, and Google Cloud. Users can now access computing resources on demand without owning the physical infrastructure.

Thus, both hardware improvements and Internet-based software technologies played an important role in the development of cloud computing.

Explain the role of server virtualization in cloud computing Difference b/w virtualization as an enabling technology and cloud computing as a service model.

Server virtualization is the process of creating mutiple virtual machines on a single physical server using a hypervisor. Each VM can run its own operating system and applications.

Role of virtualization in cloud computing:-

- Better resource utilization:- multiple VMs can share one physical server.
- Isolation:- Each VM operates independently from other VMs.
- scalability:- VMs can be quickly created, removed or resized.
- cost reduction:- Fewer physical servers are required.
- Flexibility:- virtual machines can be easily managed and moved.

virtualization	cloud computing
• Technology for creating virtual resources.	• service model for delivering computing resources.
• Focuses on resource sharing and isolation.	• Focuses on on-demand access and scalability.
• can be used in private data centers	• can be public, private, or hybrid.
• Example:- VMware ESXi	• Example:- Amazon EC2

Therefore, virtualization is an enabling technology, while cloud computing is a service model that uses virtualization along with networking, storage, automation, and other technologies.

4) Discuss the role of data centers in delivering cloud services.

A data center is a physical facility containing servers, storage systems, networking equipment, power, cooling, and security infrastructure. It provides the physical foundation for cloud services.

Major roles of data centers:-

1. computing:- servers provide processing power for cloud applications and virtual machines.
2. Storage:- storage systems keep user files, databases, application data and backups.
3. Networking:- Networking equipment connects cloud servers with users and other services through the internet.
4. virtualization:- physical servers are divided into multiple virtual machines so resources can be shared efficiently.
5. Reliability:- Back up power, redundant servers, storage and network connections help reduce service interruptions.
6. Security:- physical security, access control, monitoring, firewalls and encryptions help protect cloud infrastructure and data.
7. Scalability:- Additional servers and storage can be added when demand increases.

Example:- AWS, Microsft Azure, and Google Cloud use large data centers in different regions to deliver services to customers.

Thus, data centers provide the physical infrastructure required to deliver reliable, secure and scalable services to customers.

5) Differentiate Iaas, Paas, Saas, and Xaas with one suitable real world example for each model.

Model	Full Form	main service	Example
Iaas	Infrastructure as a service	servers, storage, networking	Amazon EC2
Paas	platform as a service	Application development platform	Google App Engine
Saas	software as a service	complete software application	Gmail
Xaas	Anything as a service	various IT services	AWS Lambda

Iaas:-

Iaas provides basic infrastructure such as virtual machines, storage and networking. The user manages the operating system and applications.

Example:- Amazon EC2 provides virtual servers that users can configure according to their needs.

Paas:-

Paas providing a ready-made platform for developing and deploying applications. The provider manages the underlying

infrastructure.

Example:- Google App Engine allows developers to deploy applications without managing servers.

SaaS:-

SaaS provides complete software through the internet. The provider manages the application and infrastructure

Example:- Gmail allows users to access email through a web browser without maintaining an email server.

XaaS:-

XaaS means Anything as a service and refers to different IT services delivered through the Internet, such as Database as a service or Function as a service.

Example:- AWS Lambda provides Function as a service.

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